

STRATEGIC ANALYSIS REPORT

Smart Energy Consumption & Forecasting

Converting Smart Meter Data into Household Savings, Grid Stability, and Provider Forecasts

Prepared by	Natender
Program	MSc. Data Science
Institution	University of Europe for Applied Sciences, Potsdam, Germany
Academic Supervisor	Professor Humera Noor Minhas
Report Date	June 22, 2026
Presentation Date	July 6, 2026

Table of Contents

Table of Contents	2
1. Executive Summary	3
Key Outcomes at a Glance.....	3
2. Business Context & Opportunity	3
Data Foundation	4
3. Methodology & Approach.....	4
Data Quality Controls	4
Technology Stack.....	5
4. Key Findings & Analysis	5
4.1 Daily Consumption Pattern	5
4.2 Seasonal Trends	5
4.3 Weather Correlations	5
4.4 Weekday vs. Weekend Behaviour	6
5. Forecast Model & Results.....	6
5.1 Why Facebook Prophet	6
5.2 Six-Month Consumption Forecast (Sep 2023 – Feb 2024).....	6
6. Strategic Recommendations	6
6.1 Load-Shifting Programme	6
6.2 Recommendation Summary by Stakeholder	7
7. Future Roadmap.....	7
8. Project Timeline & Governance.....	7
9. Conclusion	8
Appendix: Data Sources & References	8

1. Executive Summary

Smart meters across Europe generate enormous volumes of granular consumption data, yet the overwhelming majority of this data is never converted into decisions. This project set out to close that gap: transforming sixteen months of 15-minute interval smart meter readings from 172 households in Loureiro, Portugal into a working, end-to-end energy intelligence system — one that speaks to three distinct audiences at once.

The engagement combined data engineering, exploratory analysis, and machine learning forecasting to deliver three commercial outcomes: (1) a clear off-peak load-shifting recommendation that lets households reduce electricity bills, (2) a demand profile that helps grid operators anticipate and manage peak-hour stress, and (3) a validated six-month consumption forecast that gives energy providers a defensible basis for resource and procurement planning.

Bottom line: the data shows a highly predictable, repeatable demand pattern — a dominant 20:00 evening peak and a shallow overnight baseload — that can be actively managed through a simple off-peak incentive scheme, at negligible additional infrastructure cost.

Key Outcomes at a Glance

Stakeholder	Deliverable	Value Created
Households	Personalised off-peak load-shifting windows (1–6 AM)	Lower bills via off-peak tariffs
Grid Operators	Hourly, weekly and seasonal demand profile	Reduced blackout / overload risk
Energy Providers	6-month Prophet consumption forecast (Sep 2023–Feb 2024)	More accurate resource & procurement planning

2. Business Context & Opportunity

Three stakeholder groups are currently underserved by the raw data smart meters already collect:

Stakeholder	Pain Point	Business Impact
Households	No guidance on when to run heavy appliances cheaply	Overpaying on electricity bills
Grid Operators	Demand spikes are hard to predict and manage	Risk of blackouts and overloads
Energy Providers	No accurate consumption forecast for planning	Inefficient resource allocation

Data Foundation

- 172 households in Loureiro, Portugal
- 15-minute interval smart meter readings
- 16+ months of data (May 2022 – September 2023)
- Paired with local weather data (temperature, humidity, wind, radiation)
- Source: Monteiro et al. (2024), Mendeley Data — DOI: 10.17632/vryvyfz2tj.1

3. Methodology & Approach

The project was delivered through a five-stage data pipeline, executed across five two-week sprints. Each stage built directly on the last, ensuring the forecasting model was trained on a fully validated, analysis-ready dataset rather than raw, noisy inputs.

Stage	What Was Done	Sprint
1. Data Collection	Acquired energy + weather CSVs; explored structure, time range, and 172 building meter readings	Sprint 1
2. Data Preprocessing	Time-based interpolation → forward fill → backward fill for missing values. Dropped near-empty meters (83, 87, 92, 113, 137, 144). Removed low-relevance weather columns	Sprint 1
3. Dataset Merge	Joined energy and weather data on the shared Time column into one unified dataset	Sprint 1
4. Exploratory Analysis	Identified the double-peak daily pattern, seasonality, weekday/weekend shifts, and weather correlations	Sprint 2 & 3
5. Forecasting Model	Facebook Prophet model trained on the cleaned time series; produced a 6-month forecast	Sprint 4 & 5

Data Quality Controls

Both the energy and weather datasets required cleaning before they could support reliable analysis. A consistent, time-aware approach was used throughout:

- Time-based interpolation applied first, since gaps in a time series are best estimated from neighbouring timestamps rather than a global average
- Forward fill (ffill) and backward fill (bfill) used to resolve any values interpolation could not reach
- Energy meters with more than 99% zero or null readings were dropped entirely (Meters 83, 87, 92, 113, 137, 144) as statistically non-informative
- Redundant or low-relevance weather columns were removed (Max_Inst_Wind_Speed, Inst_Temp, Avg_Wind_Direction) to reduce noise
- Energy and weather datasets were merged on the shared Time column into a single analysis-ready file

Technology Stack

Language	Data Handling	Forecasting	Visualization
Python 3	Pandas · NumPy	Facebook Prophet	Matplotlib · Power BI / Streamlit

4. Key Findings & Analysis

4.1 Daily Consumption Pattern

Consumption follows a clear, repeatable double-peak pattern every day, driven by household behaviour rather than external conditions:

Time Window	Avg. Units	Pattern
03:00–05:00	12.43	Baseload minimum
06:00–12:00	17.27	Morning rise
20:00	21.58	Primary peak

The dominant evening peak at 20:00 is driven by post-work household activity — lighting, cooking, heating/cooling, and appliance use — and represents the single highest-value target for demand-response intervention.

4.2 Seasonal Trends

Month	Avg. Units	Driver
January 2023	18.96	Heating (winter peak)
February 2023	18.07	Heating
July 2022	18.29	Cooling (air conditioning)
September 2023	13.74	Shoulder-season low

4.3 Weather Correlations

Environmental factors play a smaller role in daily consumption than expected — behaviour, not weather, is the dominant driver:

Weather Variable	Correlation	Insight
Relative Humidity	−0.10 (strongest)	High humidity days tend to be mild/rainy, reducing AC and heating demand
Wind Speed	+0.04	Windier days are weakly associated with higher heating demand

Weather Variable	Correlation	Insight
Average Temperature	≈ 0.00	Weak direct correlation — behaviour dominates over temperature
Global Radiation	≈ 0.01	Minimal direct effect on daily consumption fluctuations

4.4 Weekday vs. Weekend Behaviour

	Morning (10–12)	Evening Peak (19–21)	Key Takeaway
Weekdays	Moderate rise	Very high (post-work rush)	Concentrate demand-response here
Weekends	Earlier, higher	Lower than weekdays	More distributed load — less critical

5. Forecast Model & Results

5.1 Why Facebook Prophet

Standard ARIMA models struggle with the overlapping seasonalities present in this dataset. Prophet natively handles daily (24-hour), weekly (7-day) and yearly (365-day) cycles simultaneously — matching the double-peak, weekday/weekend, and winter/summer patterns identified during exploratory analysis. It is also robust to missing data and produces interpretable components that are easy to explain to non-technical stakeholders such as grid operators and energy providers, and it supports the future addition of weather regressors to sharpen forecasts further.

5.2 Six-Month Consumption Forecast (Sep 2023 – Feb 2024)

Month	Forecast (Total Units)	Trend
Sep 2023	41,254	Shoulder season low
Oct 2023	46,088	Autumn rise
Nov 2023	45,106	Stable
Dec 2023	47,122	Winter build-up
Jan 2024	47,563	Peak winter demand
Feb 2024	44,943	Post-peak drop

6. Strategic Recommendations

6.1 Load-Shifting Programme

The clearest, lowest-cost intervention available from this analysis is a simple off-peak scheduling programme, splitting the household base into two overnight windows to smooth grid load rather than concentrating it in a single window:

Group	Optimal Window	Appliances to Shift	Saving
50% of households	1 AM – 3 AM	EV charging, dishwashers, washing machines	Off-peak tariff
Other 50%	3 AM – 6 AM	Heavy appliances, tumble dryers	Off-peak tariff
Key benefit: spreading heavy loads across 1–6 AM reduces grid stress during the 19:00–21:00 peak, lowers consumer bills through off-peak tariff savings, and prevents infrastructure overload — a coordinated win for households, operators, and providers.			

6.2 Recommendation Summary by Stakeholder

- Households: adopt the assigned overnight window for EV charging, dishwashers, washing machines, and tumble dryers to access off-peak tariff savings.
- Grid Operators: use the hourly and weekday/weekend demand profile to plan reserve capacity around the 19:00–21:00 peak specifically, rather than uniformly across the day.
- Energy Providers: incorporate the 6-month Prophet forecast directly into procurement and resource-allocation cycles, with particular attention to the January 2024 peak.

7. Future Roadmap

#	Enhancement	Value
1	Real-Time Weather API Integration — connect live weather feeds to update forecasts dynamically as conditions change	Higher forecast accuracy; automatic re-prediction
2	Expand to Commercial & Industrial — apply the same model to larger consumption sectors for far greater cost savings	Higher business impact; grid-wide scalability
3	Smart Home Automation (IoT) — integrate recommendations with smart home devices to automatically schedule appliances	Zero manual effort; guaranteed cost savings

8. Project Timeline & Governance

The project was delivered on schedule across five sprints, each with a clearly defined focus and deliverable:

Sprint	Dates	Focus	Deliverable	Status
1	Apr 7–20	Data Collection & Cleaning	Cleaned dataset + report	Completed
2	Apr 21–May 4	EDA – Patterns	Consumption analysis report	Completed
3	May 5–18	EDA – Modelling Plan	Prophet justification report	Completed
4	May 19–Jun 1	Forecasting Model	Trained Prophet model	Completed
5	Jun 2–15	Insights & Dashboard	Recommendations + forecast	Completed

9. Conclusion

This project transformed sixteen months of raw smart meter data from 172 households into a complete, end-to-end energy intelligence system — from data cleaning and exploratory analysis through to a validated six-month demand forecast and concrete, actionable load-shifting recommendations.

The result is a repeatable framework that delivers value to all three stakeholder groups simultaneously: smarter, lower-cost households; more resilient, better-planned grids; and energy providers equipped with a defensible, data-driven forecast. The roadmap items outlined in Section 7 provide a clear path to scale this framework beyond a single residential dataset into a broader commercial energy-intelligence capability.

Appendix: Data Sources & References

- Dataset: Monteiro et al. (2024), Mendeley Data — DOI: 10.17632/vryvyfz2tj.1
- Institution: University of Europe for Applied Sciences, Potsdam, Germany
- Academic Supervisor: Professor Humera Noor Minhas
- Source sprint reports: Sprint 1 (Data Cleaning & Preprocessing, Apr 19 2026), Sprint 2 (Energy Consumption Analysis, May 2 2026), Sprint 3 (Exploratory Data Analysis & Forecasting Plan, May 18 2026), Final Capstone Report (Jun 22 2026)